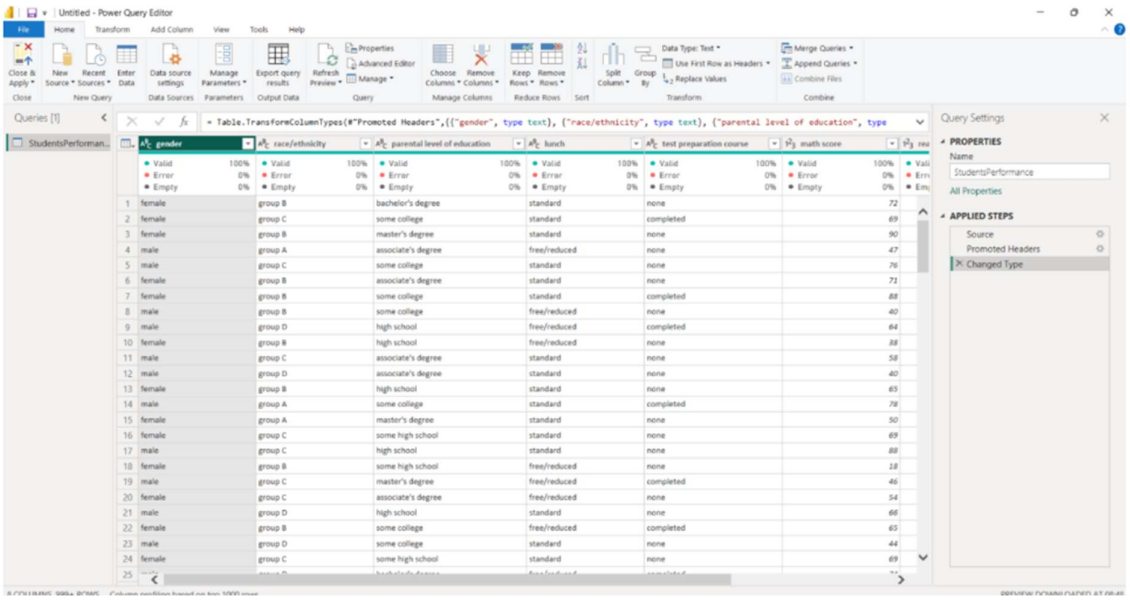
POWERBI WEEK 7 REPORT

NAME: DHARSHANA P A

DEPT: BSC DS AND AI 2ND YEAR

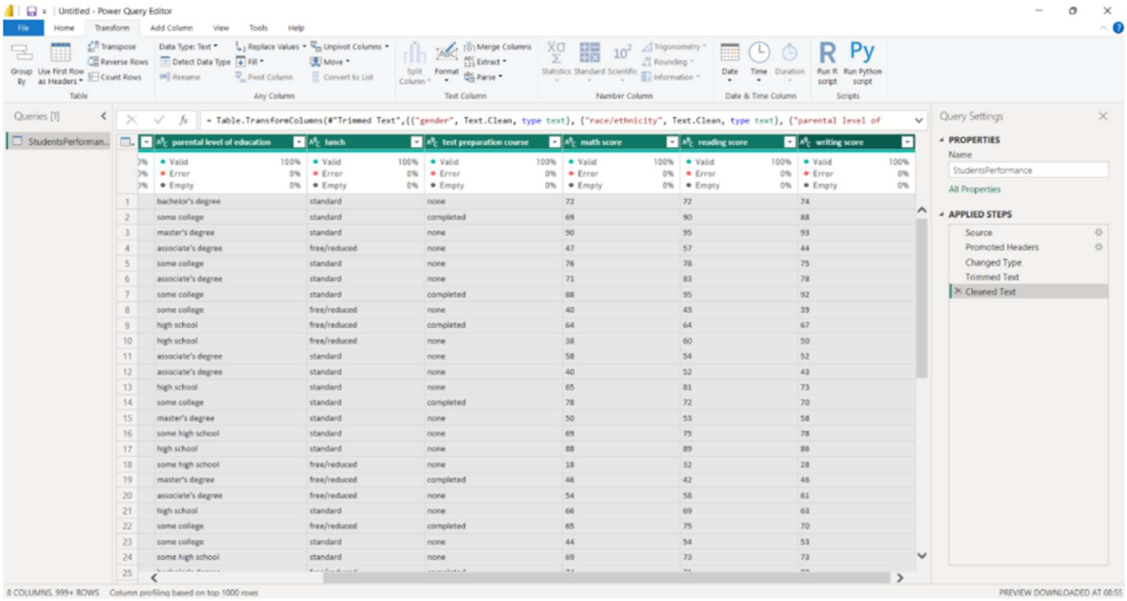
IMPORTING THE STUDENT PERFORMANCE DATASET

THE STUDENT PERFORMANCE DATASET WAS IMPORTED INTO POWER BI AND OPENED IN POWER QUERY. THE DATASET CONTAINS STUDENT INFORMATION ALONG WITH THEIR MATH, READING AND WRITING SCORES.



CLEANING TEXT DATA

THE TEXT COLUMNS WERE CLEANED USING THE CLEAN TRANSFORMATION. THIS WAS APPLIED TO REMOVE UNWANTED CHARACTERS AND ENSURE THAT THE CATEGORICAL VALUES WERE CONSISTENT.



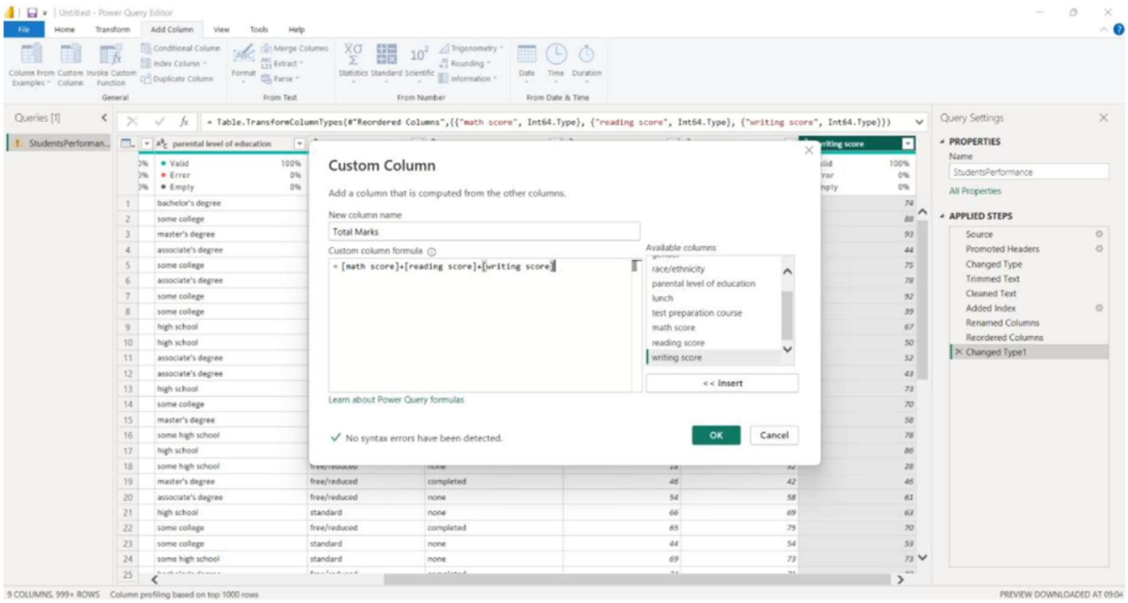
CREATING TOTAL MARKS

A CUSTOM COLUMN NAMED TOTAL MARKS WAS CREATED BY ADDING THE THREE SUBJECT SCORES.

FORMULA:

[MATH SCORE] + [READING SCORE] + [WRITING SCORE]

THIS GIVES THE TOTAL MARKS OBTAINED BY EACH STUDENT ACROSS THE THREE SUBJECTS.



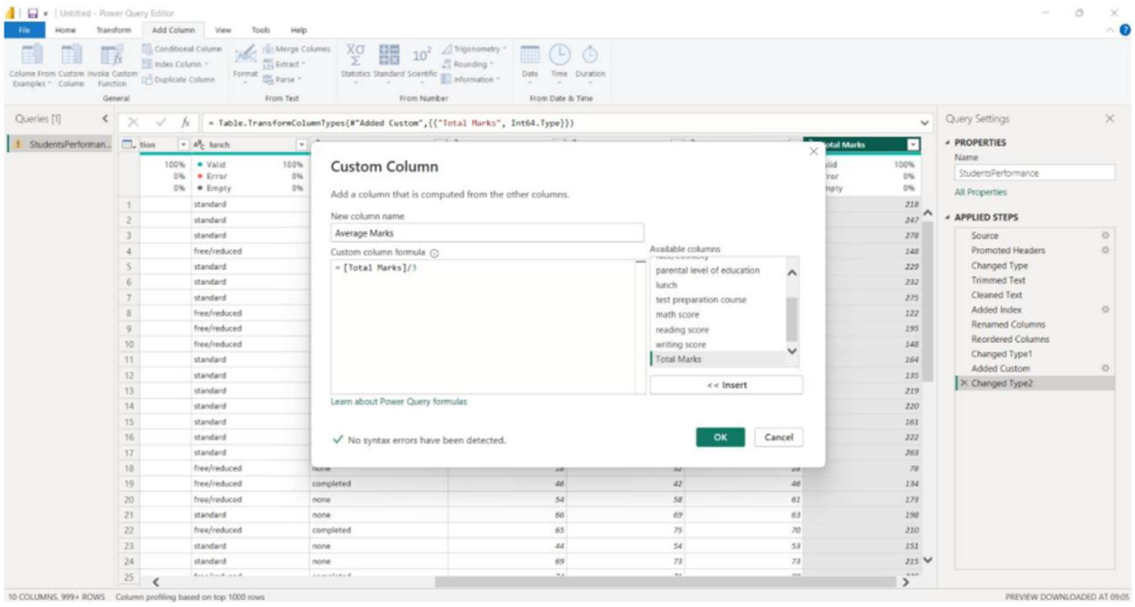
CREATING AVERAGE MARKS

A CUSTOM COLUMN NAMED AVERAGE MARKS WAS CREATED BY DIVIDING THE TOTAL MARKS BY 3.

FORMULA:

[TOTAL MARKS] / 3

THIS CALCULATES THE AVERAGE SCORE OF EACH STUDENT ACROSS THE THREE SUBJECTS.



AVERAGE MARKS COLUMN

THE AVERAGE MARKS COLUMN WAS ADDED TO THE DATASET. THE VALUES REPRESENT THE AVERAGE PERFORMANCE OF EACH STUDENT ACROSS MATH, READING AND WRITING.

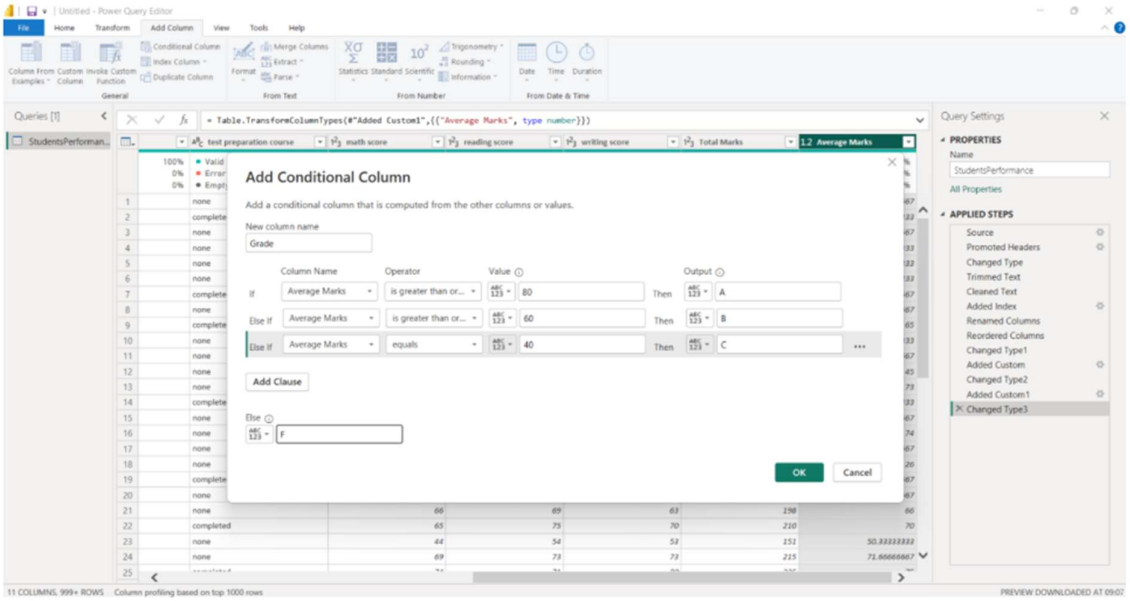
CREATING GRADE

A CONDITIONAL COLUMN NAMED GRADE WAS CREATED BASED ON AVERAGE MARKS.

GRADING CONDITIONS:

- AVERAGE MARKS ≥ 80 → A
- AVERAGE MARKS ≥ 60 → B
- AVERAGE MARKS = 40 → C
- OTHERWISE → F

THIS COLUMN CATEGORIZES STUDENTS BASED ON THEIR AVERAGE PERFORMANCE.



CREATING PASS/FAIL RESULT

A CONDITIONAL COLUMN NAMED RESULT WAS CREATED USING AVERAGE MARKS.

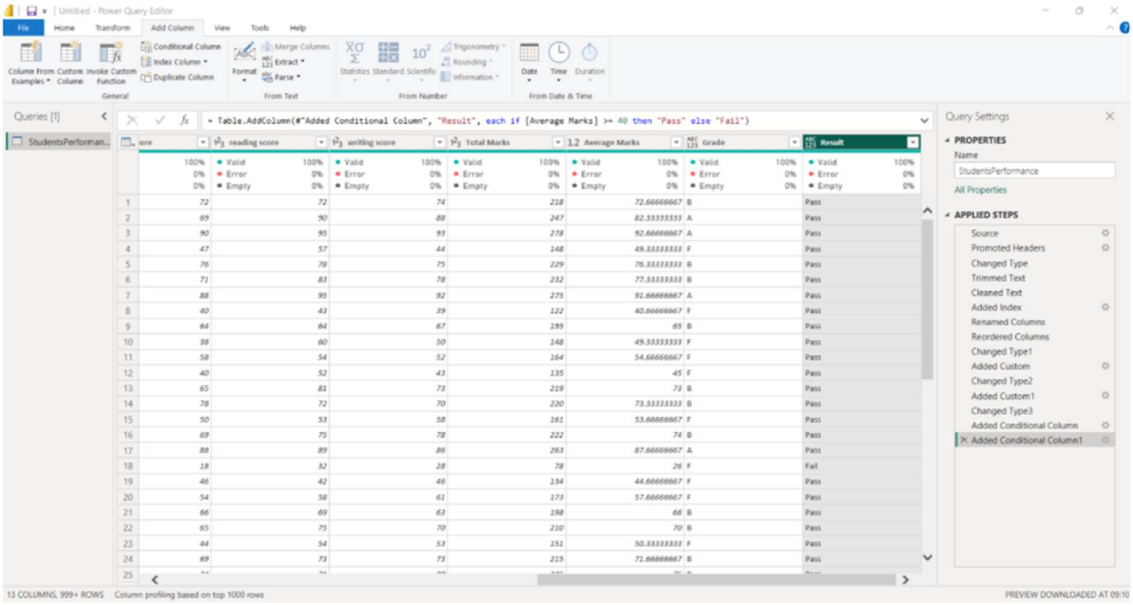
CONDITION:

- AVERAGE MARKS ≥ 40 → PASS
- OTHERWISE → FAIL

THIS WAS USED TO IDENTIFY WHETHER EACH STUDENT MET THE MINIMUM PASSING REQUIREMENT.

RESULT COLUMN

THE RESULT COLUMN WAS ADDED SUCCESSFULLY. STUDENTS WITH AN AVERAGE MARKS VALUE OF 40 OR ABOVE ARE MARKED AS PASS, WHILE THE REMAINING STUDENTS ARE MARKED AS FAIL.



UNPIVOTING SCORE COLUMNS

THE THREE SUBJECT SCORE COLUMNS WERE SELECTED AND THE UNPIVOT COLUMNS OPTION WAS USED. THIS CONVERTS THE SEPARATE SUBJECT COLUMNS INTO ROWS, MAKING THE DATA EASIER TO ANALYSE BY SUBJECT.

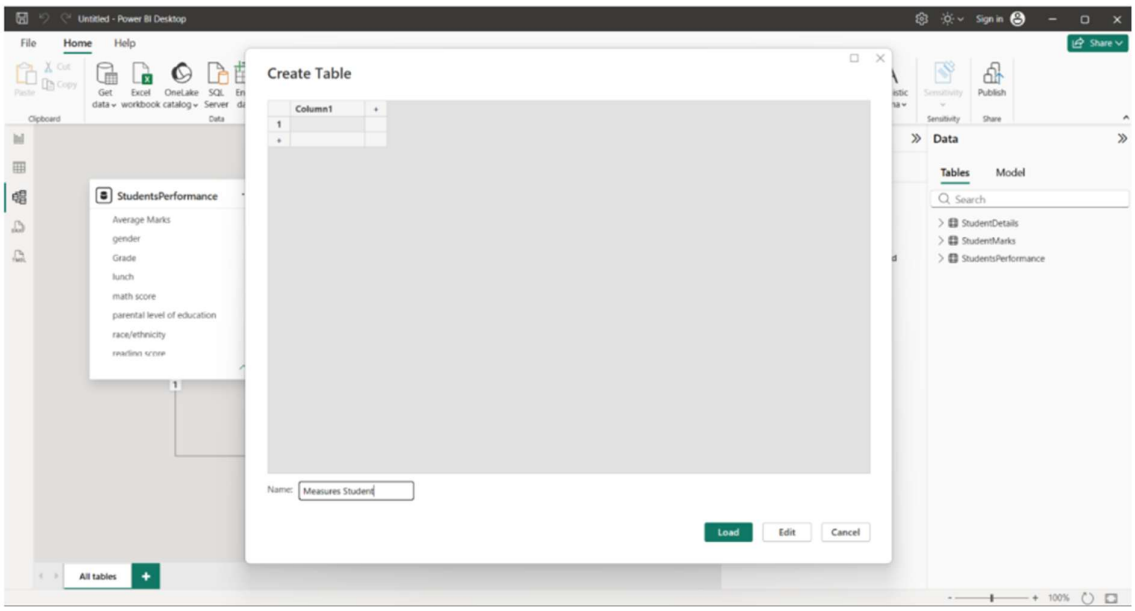
RENAMING UNPIVOTED COLUMNS

AFTER UNPIVOTING, THE ATTRIBUTE COLUMN WAS RENAMED TO SUBJECT AND THE VALUE COLUMN WAS RENAMED TO SCORE.

THE RESULTING STUDENTMARKS TABLE CONTAINS:

- STUDENT ID
- SUBJECT
- SCORE

THIS CREATES A MORE FLEXIBLE STRUCTURE FOR ANALYSING SCORES ACROSS DIFFERENT SUBJECTS.



AVERAGE MATH SCORE

THE AVERAGE MATH SCORE MEASURE WAS CREATED USING THE AVERAGE FUNCTION.

DAX FORMULA:

AVERAGE MATH SCORE =

AVERAGE(STUDENTSPERFORMANCE[MATH SCORE])

THIS MEASURE CALCULATES THE AVERAGE MATH SCORE OF THE STUDENTS.

AVERAGE READING SCORE

THE AVERAGE READING SCORE MEASURE WAS CREATED USING THE AVERAGE FUNCTION.

DAX FORMULA:

AVERAGE READING SCORE =

AVERAGE(STUDENTSPERFORMANCE[READING SCORE])

THIS MEASURE CALCULATES THE AVERAGE READING SCORE OF THE STUDENTS.

AVERAGE WRITING SCORE

THE AVERAGE WRITING SCORE MEASURE WAS CREATED USING THE AVERAGE FUNCTION.

DAX FORMULA:

AVERAGE WRITING SCORE =

AVERAGE(STUDENTSPERFORMANCE[WRITING SCORE])

THIS MEASURE CALCULATES THE AVERAGE WRITING SCORE OF THE STUDENTS.

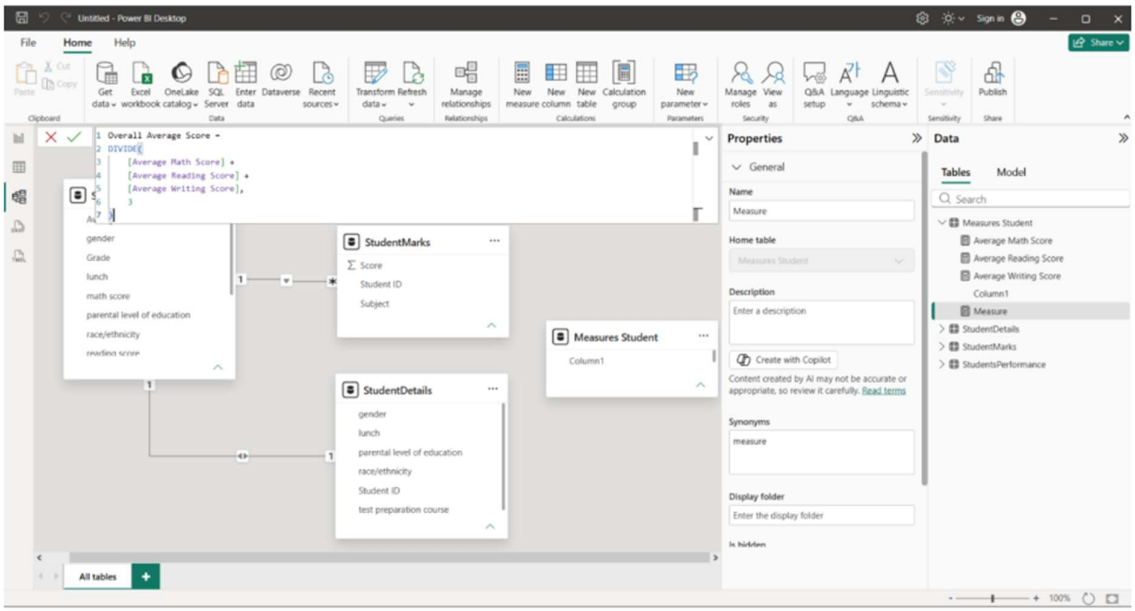
OVERALL AVERAGE SCORE

THE OVERALL AVERAGE SCORE MEASURE WAS CREATED BY TAKING THE AVERAGE OF THE THREE SUBJECT-LEVEL AVERAGE SCORES.

DAX FORMULA:

```
OVERALL AVERAGE SCORE =
DIVIDE(
    [AVERAGE MATH SCORE] +
    [AVERAGE READING SCORE] +
    [AVERAGE WRITING SCORE],
    3
)
```

THIS PROVIDES AN OVERALL AVERAGE SCORE ACROSS MATH, READING AND WRITING.



FINAL DATA MODEL

THE FINAL POWER BI MODEL CONTAINS THE MAIN STUDENTSPERFORMANCE TABLE ALONG WITH THE STUDENTMARKS, STUDENTDETAILS AND MEASURES STUDENT TABLES.

THE STUDENTMARKS TABLE CONTAINS THE SUBJECT-WISE SCORE INFORMATION, WHILE STUDENTDETAILS CONTAINS THE MAIN STUDENT INFORMATION. THE MEASURES STUDENT TABLE STORES THE DAX MEASURES CREATED FOR ANALYSIS.

THE FINAL MODEL PROVIDES A STRUCTURED WAY TO ANALYSE STUDENT PERFORMANCE AND CALCULATE OVERALL ACADEMIC METRICS.

